Venkata Abhishek Gullipalli

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Summary

Innovative AI/ML Engineer with 4+ years of experience in building scalable machine learning and NLP solutions across healthcare, CRM, and research domains. Skilled in designing end-to-end ML pipelines, RAG-based conversational agents, predictive analytics, and real-time cloud deployments using AWS, Azure, and GCP. Experienced in model optimization, explainable AI, and integrating AI with enterprise workflows to improve efficiency, automate decision-making, and drive business and operational impact.

Technical Skills

- **Programming & Data Engineering**: Python (PyMuPDF, Pydantic, pandas, spaCy, PyTorch, TensorFlow, scikit-learn, XGBoost, LSTM, ARIMA, Prophet), SQL, PostgreSQL, MySQL, Hive, PySpark, ETL workflows, data ingestion, schema design, feature engineering, Airflow, real-time data pipelines, data drift monitoring.
- **Machine Learning & NLP**: Hugging Face Transformers (DeBERTa-V3, Longformer), LoRA, PEFT, Optuna, MLflow, LangChain, hyperparameter tuning, Bayesian optimization, k-fold cross-validation, reinforcement learning, few-shot & zero-shot learning, self-supervised learning, explainable AI (SHAP, XAI), RAG, synthetic data generation, AutoML, NAS.
- **AI & Conversational Agents**: RAG architecture, LangChain, FAISS, ChromaDB, ViT, GPT-4, PubMed search integration, AI chatbots, speech-to-text, contextual question answering, medical research agents, real-time SOAP note generation, LangGraph
- **Cloud, Deployment & Infrastructure**: AWS, Azure, GCP, Docker, Kubernetes, REST APIs, serverless architecture (AWS Lambda, Azure Functions), Kafka, cloud deployment, model monitoring (Prometheus, Grafana), Streamlit, Plotly, scalable low-latency inference pipelines.
- **Applied Math & Statistics:** Probability theory, Bayesian inference, hypothesis testing, causal inference (propensity, DAG reasoning), time-series modeling, Markov processes, convex / non-convex optimization, Bayesian hyperparameter optimization (Optuna), simulation-based evaluation.
- **Compliance, Collaboration & Soft Skills**: HIPAA, SOC 2, IT risk controls, stakeholder communication, requirement gathering, ethical AI practices.

Professional Experience

AI Engineer, ECare Medical Group | 06/2025 – Present | TX, USA

- Established an AI evaluation framework using Ragas and LLM-as-a-judge methods, improving faithfulness and relevance benchmarks, achieving 92% accuracy, reducing hallucinations by 15%, and strengthening automated medical query validation through continuous testing pipelines.
- Implemented Responsible AI and safety protocols, including crisis detection escalation pathways, anti-empathy testing, and Human-in-the-Loop checkpoints, reducing unsafe responses by 24%, ensuring 99% compliance adherence, and improving trust in high-risk medical interactions.
- Engineered an advanced RAG pipeline using hybrid search combining BM25 and vector embeddings with Reciprocal Rank Fusion, enhanced by Azure Semantic Ranker, improving retrieval precision by 8% and reducing irrelevant results.
- Architected a high-precision RAG ingestion pipeline using FastAPI, ensuring 1:1 page mapping and 100% metadata accuracy across medical manuals, reducing ingestion errors by 23% and improving source attribution reliability for downstream retrieval.
- Developed an agentic orchestrator to structure LLM outputs into JSON for deep-linked Quick Review features, optimized Azure AI Search with custom embedding batching, and built scalable ingestion APIs with automated evaluation and validation suites.

AI Research Assistant, University of Houston | 05/2024 – 02/2025 | TX, USA

- Architected “NeuroChat,” a context-aware conversational system combining hybrid reasoning pipelines, semantic retrieval, and adaptive response generation, improving multi-turn dialogue coherence by 28%, boosting contextual accuracy by 22%, and enhancing user engagement metrics.
- Developed a context-persistent chatbot framework using vector embeddings, semantic re-ranking, and session memory, increasing document retrieval precision by 31%, reducing irrelevant responses and improving overall answer consistency across large-scale technical knowledge bases.
- Engineered scalable, low-latency inference pipelines using asynchronous processing, intelligent caching, and model optimization techniques, reducing response latency by 5%, improving throughput by and ensuring reliable real-time performance under high concurrent user workloads.

AI/ML Engineer, Salesforce India | 06/2021 – 12/2023 | Remote, Andhra Pradesh, India

- Architected a predictive CRM analytics platform integrating Salesforce Sales Cloud, Service Cloud, and external marketing data into Snowflake, enabling real-time lead scoring, opportunity forecasting, anomaly detection, and automated dashboards for executive decision-making.
- Built scalable data pipelines using Apache NiFi, Kafka, and Azure Data Factory, streaming customer interactions, engagement events, and sales metrics from Salesforce and third-party systems, improving data availability and predictive insights for sales teams by 18%.
- Engineered feature extraction workflows with PySpark, dbt, and embedded Tableau/Grafana metrics to capture customer behavior, deal progression, and engagement patterns, feeding ML models that improved lead conversion predictions by 12% and churn detection by 10%.
- Developed machine learning models using XGBoost, Elastic Net, and Transformers to forecast pipeline revenue, detect churn risk, and optimize cross-sell opportunities, reducing forecast error by 5% and improving upsell recommendations across global client accounts.
- Enhanced model explainability using SHAP, tuned with Optuna, validated with rolling time-series methods (RMSE 5.9, AUC 0.91), and deployed via Azure ML, enabling transparent insights for marketing campaigns and sales engagement strategies.
- Deployed predictive services using Docker, Kubernetes, KServe, and Azure Functions for serverless triggers, automating retraining workflows via Airflow and monitoring with Evidently AI, generating real-time alerts and improving actionable recommendations by 21%.

Education

Master of Science (MS) in Engineering Data Science **(GPA: 3.9/4)** | 01/2024 – 12/2025
University of Houston, Houston, TX, USA
Bachelor of Technology (B.Tech) in Computer Science and Engineering **(GPA: 9.01/10)**
GITAM University, Visakhapatnam, India

Certificates & Publications

- **Academy Accreditation - Generative AI Fundamentals**
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional**
- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- **AI Agents Fundamental**
- **NVIDIA Deep Learning**
- **Neural Networks and Deep Learning**
- **Transfer Learning for NLP with TensorFlow Hub**
- **An Experimental Study on Denoising the Images with Autoencoders**